Band-1 Hadamard Order 10040 – Classical Paley-I (q=10039); IN SPK via Paley-I; Gap 2052 Still OPEN

H_10040 Paley-I PASS; Paley-II FAIL/SKIP (q=5019=3 7239 NOT prime / not prime power; ===3 mod 4); Kronecker FAIL/SKIP (4*2510; 2510===2 mod 4 NOT Hadamard-admissible); Miyamoto digression skipped; gap 2052 OPEN; conjecture OPEN; GATE-04 LOCKED

C.M. Baird; ZoraASI mission-control Band-1 dash

2026-09-23

Document class. Scientific computational / construction certificate (Band-1).
Program. Open-mathematics collapse list * Human+AI attack order * ZoraASI mission-control.
Parents. `2026-09-23_Band1_Hadamard_10008_PaleyI`; `2026-09-23_Band1_Hadamard_9968_PaleyI`; `2026-09-23_Band1_Hadamard_9932_PaleyI`.
Lab. `/workspace/experiments/band1/hadamard_10040_paleyI/`.
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Honesty stamp (non-negotiable).
This memo claims **finite construction of order 10040 only (verified max|HHˆT-nI|=0), via classical Paley-I q=10039 (prime ===3 mod 4). 10040 IN SPK via Paley-I.** Honest skip: Miyamoto digression this wave (prefer classical SPK). Classical SPK for 10040: Paley-I **PASS** ($q = 10039$ prime $\equiv 3 \pmod 4$); Paley-II **FAIL/SKIP** ($q = 5019 = 3 \cdot 7 \cdot 239$ NOT prime / not prime power; $\equiv 3 \pmod 4$); Kronecker **FAIL/SKIP** (note $10040 = 4 \cdot 2510$; $2510 \equiv 2 \pmod 4$ NOT Hadamard-admissible). **10040** folds into SPK **only** via classical Paley-I (this run). It **does not** claim the full Hadamard conjecture. GATE-04 collection remains **LOCKED**. No RH / P≠NP / YM / Frankl / NS-solved / AC-solved claims. Gap **2052** remains **OPEN**. `clay_closed = false`. Prior SPK **10008+9968+9932+9924+9908+9888+9884+9872+9860+9852+9840+9812+9804+9792+9788+976** retained; Miyamoto-only **8948/8644/8516/8452/7988/...** excluded.

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Abstract

Finite construction of order **10040**.

- H_{10040} = Paley-I ($q = 10039$) – classical SPK

- Paley-II for 10040: **FAIL/SKIP** ($q = 5019 = 3 \cdot 7 \cdot 239$ NOT prime / not prime power; $\equiv 3 \pmod 4$)
- Kronecker for 10040: **FAIL/SKIP** ($4 \cdot 2510$; $2510 \equiv 2 \pmod 4$ NOT Hadamard-admissible)
- Skip: Miyamoto digression this wave

SPK: IN SPK via Paley-I.
Gap 2052: OPEN. **Conjecture:** OPEN. **clay_closed:** false. **GATE-04:** LOCKED.

```
python3 verify_hadamard_10040.py
```

Results. overall_pass=True; units 196/196; row blocks 79/79 (block size 128); wall 852.6262s; constructed max|HHˆT-nI|=0.

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Arithmetic

- Target $n = 10040 = q + 1$ with $q = 10039$ prime $\equiv 3 \pmod 4$ — Paley-I eligible.
- Paley-II: $q = 5019 = 3 \cdot 7 \cdot 239$ NOT prime / not prime power; $\equiv 3 \pmod 4$ — FAIL/SKIP.
- Kronecker: $10040 = 4 \cdot 2510$; $2510 \equiv 2 \pmod 4$ NOT Hadamard-admissible — FAIL/SKIP.
- Miyamoto digression skipped this wave (prefer classical SPK).

Constructions (verified)

Object	Method	max|HHˆT-nI|	Wall (s)
H_10040 Paley-I	Paley-I q=10039	0	832.1347
H_10040 Paley-II	FAIL/SKIP (q=5019=3*7*239 NOT prime / not prime power; ===3 mod 4)	–	–
H_10040 Kronecker	FAIL/SKIP (4*2510; 2510===2 mod 4 NOT Hadamard-admissible)	–	–

SPK closure note

Seeds include prior SPK through **10008** lineage (**10008**; **9968**; **9932**; **9924**; **9908**; **9888**; **9884**; **9872**; **9860**; **9852**; **9840**; **9812**; **9804**; **9792**; **9788**; **9768**; **9744**; **9740**; **9720**; **9680**; **9644**; **9632**;

9624; **9620**; **9588**; **9552**; **9548**; **9540**; **9512**; **9492**; **9480**; **9468**; **9420**; **9404**; **9392**; **9372**; **9344**; **9324**; **9320**; **9312**; **9284**; **9240**; **9228**; **9204**; **9200**; **9188**; **9152**; **9104**; **9128**; **9124**; **9092**; **2220+8884**; **2236**; **2160**; **2140+8564**; . . . ; **NOT** 8948/8644/8516/. . .) + verified **10040** (+ prior **10008+9968+9932+9924+9908+9888+9884+9872+9860+9852+9840+9812+9804+9792+9788+9768**
Next constructive gap **2052** (OPEN).

Quantity	Value
10040 in SPK	**True** (Paley-I q=10039)
Gap 2052	**OPEN**
clay_closed	false
GATE-04	LOCKED

Honesty

- clay_closed = false
- GATE-04 LOCKED
- Hadamard conjecture OPEN
- Gap 2052 OPEN (method triage / non-finding only)
- No RH / P!=NP / YM / Frankl / NS / AC solved claims
- SPK membership only via classical Paley-I/II/Kronecker (not Miyamoto alone)

Next optional

q=10067->n=10068 Paley-I DIRECT q=10067 OR q=10099->n=10100 Paley-I DIRECT q=10099 OR q=10103->n=10104 Paley-I DIRECT q=10103 OR q=10111->n=10112 Paley-I DIRECT q=10111 OR Track B / Object 1 meter await